Final Project Proposal: Sight, Draft, or Swim Blind? Open-Water Race Strategy as a POMDP

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1. Goal

We want to reach a buoy in the minimum time during an open-water swim race. With no line to follow, swimmers check direction only by *sighting*, briefly lifting the eyes or head while swimming. Sighting slows a swimmer down and costs time, but skipping sighting costs distance since without a visual reference, currents push swimmers off line. Other swimmers add a second trade-off. Drafting just behind a faster swimmer allows us to hold their pace for some time and get a free directional cue, but the follower inherits the leader's navigation errors.

Sighting is an *information-gathering action* worth taking only when its value of information exceeds its time cost. We will compute a policy for one swimmer, the *agent*, that decides every few strokes whether to sight, how to steer, and whether to follow a leader, while racing a rule-following field. Success means (a) beating the best-tuned rule-based policies in an expected finish time across currents and field sizes, (b) insight into *when* sighting and drafting pay off, and (c) measuring the value of information against an oracle policy with perfect information. Our main criterion is a best response to the fixed field, but a stretch goal is a Nash equilibrium where every swimmer adapts.

2. Decision Making

Swimmers start at random beach positions with random fixed paces and race through a current to a buoy at $(L, 0)$, deciding each breathing cycle. The following model describes the open-water swimming problem as an MDP.

States $s = (x, y, \psi, \mathbf{c}, \mu, \mathbf{z}_{1:K})$: the agent's position (x, y), heading relative to the start–buoy line (ψ), the water current $(\mathbf{c})$, a mode $\mu \in \{\text{free}, \text{following}\}$; and $\mathbf{z}_{1:K}$ represents each field swimmer's position, heading, pace, and drafting mode.

Actions $a = (m, \delta, d)$: sight or not, $m \in \{0, 1\}$; turn, $\delta \in \{-\Delta, 0, +\Delta\}$, when free; and, when a swimmer is in the draft zone, follow or go solo, $d \in \{\text{follow}, \text{solo}\}$.

Transitions Heading drifts as $\psi' = \psi + \delta + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma_\psi^2)$ as a random heading noise parameter. The agent moves each decision cycle a displacement equal to $\ell(\cos\psi, \sin\psi) + \tau_m \mathbf{c} + \mathbf{w}$, where ℓ is the distance per cycle (determined by swimmer pace and drafting state), $\tau_m = \tau_0 + \tau_\kappa m$ includes the time cost τ_κ of a sight, and $\mathbf{w}$ is Gaussian noise in the position. A following agent tracks the leader's path at speed $\min(v_{\text{lead}}, (1 + \beta)v)$ with adjustable parameter β for speed boost from drafting. As a default policy for other swimmers in the field, other agents sight every N_k strokes, re-aim at the buoy, and draft any faster swimmer they contact for a fixed time.

Observations A sight returns, the buoy's relative bearing $o = \text{atan2}(-y,\, L - x) - \psi + \eta$ with $\eta \sim \mathcal{N}(0, \sigma_o^2)$ as a noisy bearing error, plus noisy swimmer positions. Otherwise, the drafting status (from proximity to the lead swimmer) is always available.

Rewards $R = -\tau_m$ per step until the buoy is reached, so maximizing expected utility minimizes expected finish time.

3. Sources of Uncertainty

We take into account multiple sources of uncertainty in the model. This includes model uncertainty in the current pattern and strength, which will affect individual agent positions throughout the race. This model uncertainty, as well as imprecise readings of location and bearing from sightings of the buoy, propagate into state uncertainty for the individual agent. The agent is not aware of precisely where it is and what its heading and velocity vector are, motivating optimization to identify strategies for when to sight. Persistent current-related drifts will also take multiple sightings of the buoy to identify and calibrate against. Finally, the strategies of other swimmers are also uncertain, which means no static optimal drafting strategy can exist. Although not in the baseline plan for this project, it would be interesting to examine multiagent game-like interactions between different agents pursuing their own drafting strategies under uncertainty about competitors' strategies.

4. Sketches of Solution

We will compare our solution to baseline strategies for open-water swimming including sighting every N strokes and re-aiming at the buoy and deterministic rules about drafting (never, always, or for a fixed amount of time). Across a range of evaluation conditions (different number of swimmers, current strength, pace distribution), we aim to identify a robust solution which consistently outperforms the best baseline strategy in each condition.

A first approach toward this solution would be online planning with Monte Carlo tree search with progressive widening, with a discretized version of the state space in position and heading state. When extending to a fully continuous version of the state and observation space, we plan on evaluating POMCPOW as an MCTS solver as described in Sunberg and Kochenderfer (2018). We will compare against baselines on the basis of time-to-objective across the full distribution of evaluation conditions over multiple rollouts, as well as total distance swam, sights taken, and time drafting.

5. AI Disclosure Statement

We used Claude Opus 5 to brainstorm potential projects before deciding on this topic.

References

Zachary N. Sunberg and Mykel J. Kochenderfer. Online algorithms for POMDPs with continuous state, action, and observation spaces. In *International Conference on Automated Planning and Scheduling (ICAPS)*, pages 259–263, 2018.